Dataset 2 – Performance improvement for models

Train dataset: dec2025rb2r2_updated.txt
Test dataset: feb2026rb2r2_updated.txt

Summary:
Several architectural and training improvements were explored across GRU, Transformer, and GAN models. While enhancements such as dropout, increased model capacity, and learning rate scheduling were applied to GRU and Transformer, they resulted in slight performance degradation, leading to conclusion that original configurations were optimal for these models. **In contrast, significant improvements were achieved in the GAN model by introducing a hybrid loss function combining adversarial and regression objectives, along with a refined training process. This transformed GAN from an ineffective model into a moderately reliable predictor.** Additionally, multiple evaluation metrics were incorporated to provide a comprehensive assessment of model performance.

In this improvement additional **evaluation metrics** were added along with the old ones:
1. MAE (Mean Absolute Error): Average error magnitude
2. RMSE (Root Mean Squared Error): Penalizes large errors more
3. R^2 (Coefficient of Determination): Range: ($-\infty$ to 1)**,** indicates how well model explains data
1 - perfect
0 - same as mean prediction
< 0 - worse than mean
4. MAPE (Mean Absolute Percentage Error): Relative error
5. Max Error: worst-case error
6. Median Absolute Error: middle error value
7. Error Std (Standard Deviation of Error): spread of errors

Comparison table:

Model	**Stage**	**MAE**	**RMSE**	**R^2**	**MAPE**	**Max Error**	**Median AE**	**Error Std**
GRU (Voltage)	Earlier	0.06499	0.09529	0.9198				
	After Changes	0.06427	0.09634	0.9180	0.1586	2.172	0.03978	0.09628
GRU (Power)	Earlier	6.2733	39.375	0.9219				
	After Changes	7.2818	41.828	0.9119	1.5900	2937.39	3.1886	41.7946
Transformer (Voltage)	Earlier	0.08255	0.10727	0.8984				
	After Changes	0.06076	0.09270	0.9241	0.1499	1.8935	0.05002	0.09266

Transformer (Power)	Earlier	23.7880	54.516	0.8503				
	After Changes	23.0626	69.445	0.7571	6.1789	2860.04	18.1023	67.4166
GAN (Voltage)	Earlier	2.8533	3.1340	-85.7157				
	After Changes	0.07915	0.12145	0.8698	0.1953	1.8117	0.06973	0.12111
GAN (Power)	Earlier	64.4984	136.831	0.0571				
	After Changes	23.5102	108.389	0.4083	5.2404	3072.46	14.2490	108.3853

The proposed improvements significantly enhance GAN performance, converting it from a failing model (negative R^2) to a moderately effective one (R^2 approx. 0.87 for voltage and approx. 0.41 for power). However, high maximum error across all models indicates limitations in handling extreme conditions.

1. GRU Improvements:

Removed aoce mode

Increased model capacity: GRU(32 → 64), added Dense layers

Added Dropout (0.2)

Added learning rate scheduler (ReduceLROnPlateau)

Increased early stopping patience (5 → 6)

Outcome:

Slight performance degradation

Original model already optimal

2. Transformer Improvements:

Removed aoce mode

Added Dropout inside transformer block

Added learning rate scheduler

Increased patience

Outcome:

Slight drop in: R^2 and RMSE performance

3. GAN Improvements:

Hybrid Loss Function

Fixed Loss Aggregation

Custom Training Loop

Generator Improvement

Generator Optimizer Fix

Outcome:

Huge improvement in R^2